

Structural bioinformatics

GOBoost: leveraging long-tail gene ontology terms for accurate protein function prediction

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Abstract

Motivation: With the advancement of deep learning, researchers have increasingly proposed computational methods based on deep learning techniques to predict protein function. However, many of these methods treat protein function prediction as a multi-label classification problem, often overlooking the long-tail distribution of functional labels (i.e., Gene Ontology Terms) in datasets. To address this issue, we propose the GOBoost method, which incorporates the proposed long-tail optimization ensemble strategy. Besides, GOBoost introduces the proposed global-local label graph module and multi-granularity focal loss function to enhance long-tail functional information, mitigate the long-tail phenomenon, and improve overall prediction accuracy.

Results: We evaluate GOBoost and other state-of-the-art (SOTA) protein function prediction methods on the PDB and AF2 datasets. The GOBoost outperformed SOTA methods on all evaluation metrics for both datasets. Notably, in the AUPR evaluation on the PDB test set, GOBoost improved by 10.71%, 35.91%, and 22.71% compared to the SOTA HEAL method in the MF, BP, and CC functions. The experimental results show the necessity and superiority of designing models from the label long-tail distribution perspective.

Availability and implementation: The source code of GOBoost is available at <https://github.com/Cao-Labs/GOBoost>.

1 Introduction

Proteins are essential molecules in biological processes, playing critical roles in catalysis, recognition, and transmission of signals, and providing structural support for cells and tissues (Giri and Cheng 2024). By advancing our knowledge of protein function, we can uncover disease mechanisms and facilitate drug development. The Gene Ontology (Huntley *et al.* 2015) Consortium has established a standardized functional vocabulary library, Gene Ontology, to define and describe specific functions. These functions are categorized into three GO categories: molecular function (MF), biological process (BP), and cellular component (CC). With advancements in protein sequencing technology, the number of protein sequences has increased dramatically. Although the traditional experimental method of annotating protein functions is accurate, it is time-consuming and labor-intensive. Therefore, there is an increasing gap between the number of surging protein sequences and the number of proteins that have been experimentally annotated (Boadu *et al.* 2025). Therefore, it is crucial to develop a fast and accurate computational method to classify protein's function type.

Knowledge transfer-based functional prediction methods (Altschul *et al.* 1990; Das *et al.* 2015; Buchfink *et al.* 2021) are

usually based on the assumption that proteins with high sequence similarity tend to exhibit similar functional properties. These methods rely on prior database knowledge, in which matching the database during large-scale data processing can be highly time-consuming. With advancements in deep learning technology, researchers are increasingly focusing on developing computational methods that address these issues. Protein sequences are the most commonly used feature information for protein function prediction. Sequence-based methods (Kulmanov *et al.* 2018; Kulmanov and Hoehndorf 2020; Cao and Shen 2021; Ranjan *et al.* 2023) aim to reveal the primary structural correlations between amino acids in proteins and mine potential evolutionary information to enhance the performance of prediction models. The latest research (Zhao *et al.* 2022; Gu *et al.* 2023; Yuan *et al.* 2023; Abdine *et al.* 2024) shows that using the pre-trained biological large language model (Elnaggar *et al.* 2021; Rives *et al.* 2021; Lin *et al.* 2022) based on the transformer architecture to extract sequence features and applying these features to downstream function prediction tasks can achieve better results.

The above sequence-based methods are easily affected by amino acid variations, while the 3D structure is often more conserved and closely related to protein function. To this end, GAT-GO (Lai and Xu 2022) uses RaptorX (Wang *et al.* 2017)

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to directly predict the amino acid contact graph in the three-dimensional structure of proteins. With the development of high-quality protein structure prediction methods (Jumper *et al.* 2021), researchers generally use the 3D structure of proteins to more accurately abstract amino acid contact graphs. On this basis, various graph convolutional networks (Kipf and Welling 2016; Lai and Xu 2022) are widely used to mine the complex relationship between protein structure information and function. Therefore, this type of methods (Gligorijević *et al.* 2021; Lai and Xu 2022; Boadu *et al.* 2023; Gu *et al.* 2023) can generally be summarized as a graph multi-label classification problem. Considering the natural characteristics of multi-label classification problems, some researchers have tried to use the information between labels to improve the prediction performance of the model. One approach, known as the hierarchical output methods (Kulmanov *et al.* 2018; Notaro *et al.* 2021), directly utilizes the hierarchical constraint information of the functional Directed Acyclic Graph (DAG) to update prediction probabilities, thereby avoiding sub-optimal solutions. Another approach does not strictly incorporate the hierarchical constraint information of the functional DAG. They usually update the functional label information by analyzing the associations between various labels. PANDA2 (Zhao *et al.* 2022) and PANDA-3D (Zhao *et al.* 2024) use the bidirectional DAG of GO terms and the GO terms relationship implicitly captured by the transformer decoder to update the label information, respectively. DeepFPF-CO (Li *et al.* 2023) uses the prior knowledge of GO terms from the training set to construct a conditional co-occurrence matrix to update the predicted probability. At the same time, DeepFPF-CO also effectively reduces the noise introduced by low-frequency terms by setting a specific threshold.

The computational methods discussed above often treat the protein function prediction problem as a straightforward multi-label classification task. Due to how the functional DAG is constructed, functions at more profound levels are usually more specific. However, only a few proteins are annotated with these specific functions. Therefore, a few terms with broad functions appear frequently in the dataset, while most terms with more specific functions appear less frequently, resulting in a right-skewed distribution with a long tail. This unbalanced distribution of GO terms is the long-tail phenomenon, as shown in [Supplementary Fig. S1](#). This unbalanced data distribution will cause the model to be more inclined to predict categories with higher frequency and ignore categories with lower frequency during training, thus affecting the overall performance of the model. Therefore, adopting a simple multi-label classification approach may cause the model to ignore these more specific low-frequency functions. To address this, Al-Shahab *et al.* (Al-Shahib *et al.* 2005) and Jaehee Jung *et al.* (Jung and Thon 2006) proposed the under-sampling technique to balance the distribution of GO terms. However, with the continuous expansion in the number of functional categories and expansion of the dataset, it is becoming increasingly difficult to mitigate the long-tail phenomenon effect using sampling technology.

When faced with a large amount of data, traditional sampling methods have difficulty dealing with the model learning bias problem caused by the imbalance of GO terms data. It leads to a decrease in the overall prediction performance of the model. To address the above problems, we proposed GOBoost, a novel protein function prediction method designed to alleviate the long-tail effect. This method

introduces an optimized ensemble strategy tailored to the long-tail distribution, integrating three complementary base models: GOBoost^{Head}, which is trained on high-frequency labels; GOBoost^{Tail}, which is trained on medium and low-frequency labels; and GOBoost^{All}, which is trained on all labels. The base model utilizes Class Activation Mapping (CAM) (Zhou *et al.* 2016) method to map protein structure features encoded by a Graph Convolutional Network (Kipf and Welling 2016) into GO term embeddings. Inspired by related works on image recognition (Ye *et al.* 2020), these GO term embeddings are passed through the proposed global label graph module to capture the co-occurrence relationships among global high-frequency functional labels. Then the proposed local individual functional label graph is constructed to [supplement](#) relationships involving low-frequency labels. A multi-granularity focal loss function is used during training, assigning a greater weight to long-tail labels, thereby encouraging the model to focus on these underrepresented labels. Ablation study results showed that GOBoost, specifically designed to address long-tail distribution, achieved higher precision accuracy and superior generalization comparing to other state-of-the-art methods.

2 Materials and methods

2.1 Datasets

This method utilizes the same Protein Data Bank (PDB) dataset as DeepFRI (Gligorijević *et al.* 2021) and the AlphaFold 2 (AF2) dataset, which was also used by HEAL (Gu *et al.* 2023). The PDB dataset consists of natural protein structures obtained through physical experiments, which are divided into training, validation, and testing sets in a sequential 8:1:1 ratio. All annotated protein function identifiers are sourced from SIFTS (2019-06-18) (Dana *et al.* 2019), which transfers functions to PDB chains according to residue mappings between UniProtKB (release 2019-06) (Boutet *et al.* 2007) and PDB (release 24.19 version) entries. The training and validation sets use both experimental (EXP) and electronically inferred GO annotations (IEA), while the test set relies solely on EXP annotations. Additionally, each protein in the PDB contains at least one experimentally determined GO term across each functional ontology. HEAL uses homologous proteins of PDB as the enhanced dataset AF2. The protein structures in this dataset were predicted by AlphaFold 2. The IC value is a measure of the specificity of a GO term. “IC > 10” indicates that the annotation is very specific, occurs less frequently, and is more difficult to be predicted by the model. Each protein was annotated with at least one GO term with an IC [Information Content (of Reference Genome Group of the Gene Ontology Consortium 2009)] > 10. Subsequently, MMseqs (Steinegger and Söding 2018) was used to cluster protein sequences with a 25% sequence similarity, resulting in 38 185 training proteins, 4 242 validation proteins, and 567 test proteins.

To improve the model's ability to dynamically learn co-occurrence relationships between labels (i.e., continuously adjust and improve its understanding of label relationships based on training data), we further processed the PDB and AF2 datasets. All training and validation sets were preprocessed according to three GO categories by removing all proteins without corresponding functions (i.e., all proteins without positive labels were excluded). To ensure a fair evaluation, the test sets of PDB and AF2 datasets are consistent

Table 1. Protein data statistics on PDB and AF2 datasets.

Dataset	Statistic	MF	BP	CC	All
PDB	Training Size	24,956(83.48%)	23,513(78.66%)	11,295(37.78%)	29,893
	Validating Size	2,749(82.75%)	2,624(78.99%)	1,299(39.10%)	3,322
	Testing size	3,414(100%)	3,414(100%)	3,414(100%)	3,414
	Number of classes	489(17.77%)	1,943(70.60%)	320(11.63%)	2,752
AF2	Training Size	33,827(88.59%)	37,263(97.59%)	32,397(84.84%)	38,185
	Validating Size	3,708(87.41%)	4,146(97.74%)	3,663(86.35%)	4,242
	Testing Size	567(100%)	567(100%)	567(100%)	567
	Number of classes	489(17.77%)	1,943(70.60%)	320(11.63%)	2,752

Note: This table shows the number of proteins in the dataset and the number of their three types of functional labels, and records their proportion in the total. The “Proportion in the total” in the “Training Size”, “Validating Size,” and “Testing Size” items indicates the ratio of the total number of proteins with at least one function of this class to the total number of proteins in these subsets. In the “Number of classes” item, the “Proportion in the total” indicates the ratio of the total number of functions of a certain class to the total number of functions of the three classes, so the sum of the three percentages equals 100%.

with DeepFRI and HEAL. Unlike the PDB test set, the AF2 test set may contain proteins without corresponding subfunctional labels, with 137, 9, and 43 proteins lacking corresponding MF, BP, and CC functions, respectively. Detailed data statistics are shown in Table 1.

2.2 Model architecture

2.2.1 Overview

The framework of GOBoost is illustrated in Fig. 1, which primarily comprises two components: the ensemble and base models. The ensemble model integrates the prediction results of three base models, and each focuses on training on different labels: Head, Tail, and All. The base model first inputs protein sequences into a pre-trained protein large language model to extract embedding matrices. It then uses AlphaFold 2 (Jumper *et al.* 2021) to predict 3D structures, which are abstracted as protein graphs. Subsequently, the protein graph and the further processed sequence embedding are embedded into a graph convolution network to derive the final graph embedding features. Then, CAM (Zhou *et al.* 2016) is used to translate the graph embedding features into GO term label embeddings. The embeddings are then fed into the global-local label graph module, which learns the co-occurrence relationships among the GO terms. Finally, the prediction results derived from the label embeddings are linearly combined with those from the protein graph embeddings to generate the final prediction probability. This probability is used to calculate the $\mathcal{L}_{mgfl}$ loss for training the model. For specific model training details, see the supplementary material section “MODEL TRAINING”.

2.2.2 Functional terms (GO terms)

Observing the frequency of GO terms in the dataset, a few terms with broad functions are annotated by many proteins, while most terms with specific functions are only annotated by a few proteins. This leads to the long-tail phenomenon: a few functions appear frequently, while most functions appear less frequently. Infrequent GO terms (i.e., more specific) have higher IC values. In the dataset, GO terms with IC greater than 10 are rare, and GO terms with IC less than 5 are common. The specificity of GO terms is expressed as Shannon Information Content (IC) (Reference Genome Group of the Gene Ontology Consortium 2009):

$$IC(GO_i) = -\log_2 \text{Prob}(GO_i) \quad (1)$$

where $\text{Prob}(GO_i)$ is the probability of observing GO_i in the UniProt-GOA database. The long-tail phenomenon causes the model to be more inclined to predict categories with

higher frequencies during training and ignore categories with lower frequencies, which affects the overall performance. Therefore, this work integrates a basic model that focuses on predicting long-tail terms and increases the weight of long-tail terms in the loss function. This makes the model pay more attention to these long-tail function terms, thereby improving the overall prediction performance of the model.

A protein usually has multiple functions, and there are natural relationships between these functions. The global functional label graph of the dataset tends to capture the co-occurrence relationship between high-frequency GO terms, but ignores the information of low-frequency GO terms. In contrast, the local functional label graph of a specific protein can well supplement the association information between low-frequency GO terms and other GO terms. We hope to use the GO terms association information captured by the global and local label maps to improve the prediction performance of the model.

2.2.3 Ensemble strategy

Due to the uneven distribution of GO terms, training a single model tends to overlook low-frequency GO terms. To address this, we proposed a long-tail optimization ensemble strategy that combines two base models, $\text{GOBoost}^{\text{Head}}$ and $\text{GOBoost}^{\text{Tail}}$, with the base model $\text{GOBoost}^{\text{All}}$. All base models share the same model architecture and training loss function and are trained on exactly the same set of sequences. The only difference is the set of training labels they are trained on. $\text{GOBoost}^{\text{All}}$ thoroughly captures relationships between all GO terms, while $\text{GOBoost}^{\text{Head}}$ and $\text{GOBoost}^{\text{Tail}}$ specialize in high-frequency GO terms and mid-to-low frequency GO term labels, respectively.

The ensemble strategy averages the overlapping prediction results from the three base models to produce the final prediction. $\text{GOBoost}^{\text{Head}}$ and $\text{GOBoost}^{\text{Tail}}$ effectively reduce the prediction error of $\text{GOBoost}^{\text{All}}$, while $\text{GOBoost}^{\text{All}}$ balances the predictions from $\text{GOBoost}^{\text{Head}}$ and $\text{GOBoost}^{\text{Tail}}$, thereby improving the robustness of the overall model, GOBoost. The probability of GO_i prediction after optimization integration is expressed as Equation 2:

$$p_i := \begin{cases} \frac{1}{2}(a_i + b_i), & i \in G^H \cap G^A, \\ \frac{1}{2}(a_i + t_i), & i \in G^T \cap G^A \end{cases} \quad (2)$$

where a_i , b_i , and t_i denote the prediction values of GO_i given by All, Head, and Tail base models. G^H is the set of

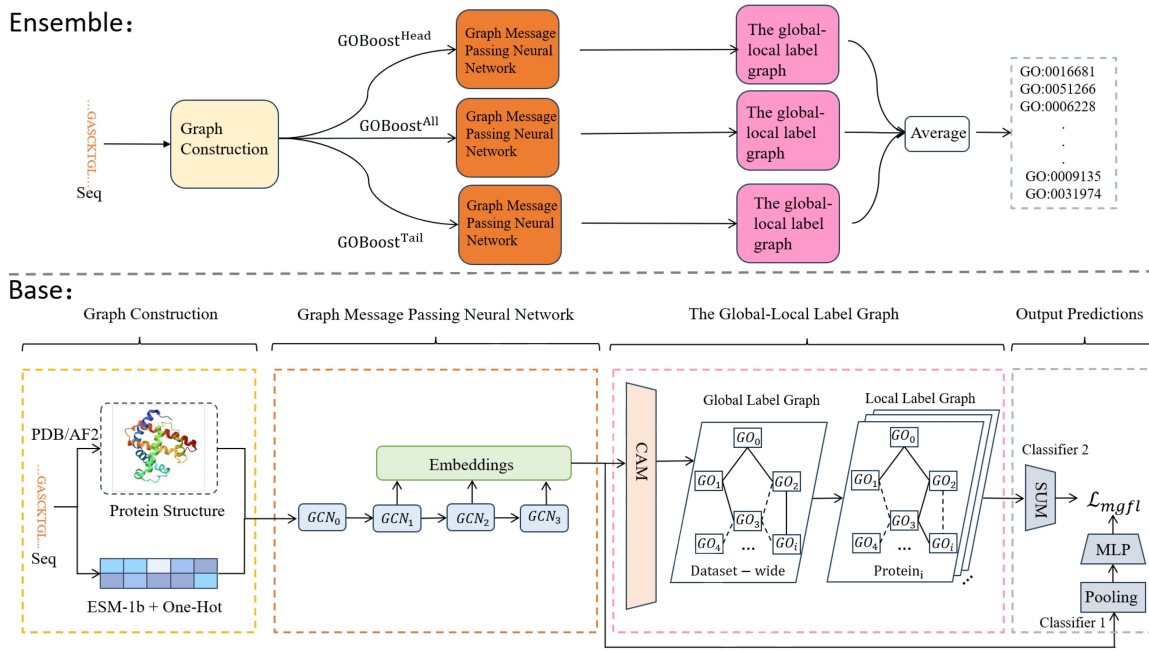


Figure 1. Overview of the proposed GOBoost. Ensemble: GOBoost is integrated from GOBoost^{All}, GOBoost^{Head}, and GOBoost^{Tail}. Base: The base model learns the structural features of proteins and the co-occurrence relationship between GO terms to predict protein function. The global label graph captures the label relationship graph within the dataset, while the local label graph captures the label relationship graph of a specific protein. Solid line indicates higher relation, and dashed line indicates lower relation of the GO terms. SUM means calculating the sum of the values of each channel of the label embedding to obtain the predicted probability, MLP means processing the label embedding through a multi-layer perceptron to obtain the predicted probability, and GCN is a graph convolutional network.

high-frequency GO term labels located in the “Head” region, G^T is the set of medium and low-frequency GO term labels located in the “Tail” region, and G^A is the set of all GO term labels.

2.2.4 Input features and graph message passing

Residue embedding. In this work, we utilized the pre-trained protein large language model ESM-1b (Rives et al. 2021) to extract sequence-level features. Due to the limitation of sequence length in ESM-1b, we cut off the first 1000 residues to represent the entire protein sequence. After encoding the amino acids into 0 and 1 codes of the corresponding classes, we are mapped to the same feature dimension as the sequence embedding extracted by ESM-1b to obtain the amino acid class embedding. Then sequence embeddings extracted by ESM-1b and amino acid class embeddings were linearly added to obtain the final protein residue level embedding $X \in \mathbb{R}^{r \times c}$.

Graph construction. We abstract the 3D structure of proteins into an undirected graph $G = (V, E, X^0)$ with self-loops, where V denotes the graph nodes, E denotes the edges connecting nodes, and X^0 denotes the initial feature matrix of the graph. Each node V_i corresponds to an amino acid, represented by the α atom of the residue. An edge E_{ij} is formed between two nodes if the Euclidean distance between their α_i and α_j atoms is less than $10 \text{ \AA}$ (represented by a 0,1 matrix $A \in \mathbb{R}^{r \times r}$ to indicate the connection relationship between residues in the protein), and $x_i^0 \in \mathbb{R}^{1 \times c}$ serves as the initial feature of node i .

Graph message passing. The protein graph’s adjacency matrix A and initial feature X are fed into a graph convolution neural network, which assigns different aggregation weights to update the features of residue i based on the degree information of adjacent residues. The receptive field range of

convolution layer features varies across different layers. As the number of convolution layers increases, the receptive field range of deeper convolution features also expands. This work performs a weighted summation of the last $L-1$ layers of protein graph features through a learnable aggregation parameter to generate the final graph feature representation.

2.2.5 Global-local label graph

Specifically, CAM (Zhou et al. 2016) is a post hoc interpretability method that is often used in the field of protein function prediction to identify functionally important residues that contribute to model predictions (Gligorijević et al. 2021; Gu et al. 2023). In the field of image recognition, the method (Ye et al. 2020) uses the importance scores of different pixels identified by CAM to aggregate and map image embeddings to label embeddings. Inspired by this, this article uses CAM as a feature conversion tool to map protein residue embeddings to label embeddings. CAM uses a classifier to identify class-specific activation scores, which are then used to weight the protein embeddings, resulting in content-aware GO term embeddings. In this way, we are able to process the relationship information between labels from the perspective of label embedding.

GOBoost uses CAM to map residue-level protein embeddings $H^f \in \mathbb{R}^{r \times c}$ to GO term embeddings $H^g \in \mathbb{R}^{g \times c}$. Unlike the static co-occurrence matrix built based on prior knowledge (Li et al. 2023), this module dynamically establishes the co-occurrence relationship between labels during training (Ye et al. 2020). The GO term embeddings are input into the global GO terms graph module. The GO term embeddings are aggregated according to the global label co-occurrence matrix, followed by the parameter matrix to update the GO term embeddings. The global label co-occurrence matrix is shared for each protein, so it tends to capture the

co-occurrence relationship between high-frequency labels within the dataset during training. However, many low-frequency GO terms still exist. Their co-occurrence relationships with other GO terms differ across proteins, making these relationships more complex for the model to capture.

Therefore, to address this, we established a local label graph. The input of the local label graph is formed by concatenating the GO term embeddings extracted by the global label graph with its global features obtained by global average pooling. This module updates the GO term embeddings using the label co-occurrence matrix and the parameter matrix. Unlike the global label co-occurrence matrix, the local individual label co-occurrence matrix is learned based on the characteristics of each protein through the parameter matrix, so each protein has a different local individual label co-occurrence matrix. This approach compensates for the low-frequency GO terms information that may be overlooked by the global label graph, effectively capturing the relationship between low-frequency GO terms and other GO terms, thus enhancing the overall predictive performance of the model.

2.2.6 Output predictions

Graph pooling classification prediction (Classifier 1). This process converts residue-level protein features into graph-level representations using average pooling operations, capturing more global information. This approach enables protein function prediction by analyzing overall trends in residue characteristics. After the graph pooling operation, the graph feature is input into the Multilayer Perceptron (MLP) layer and the linear activation function (RELU). A linear activation operation is performed after increasing the dimensionality of the graph features through the first layer of MLP. Finally, the graph features are input into the second layer of MLP and sigmoid function to obtain the predicted probabilities of all protein functions.

GO term embeddings classification prediction (Classifier 2). The global-local label mapping module generates independent GO term embeddings, enabling each to be directed into its respective classification head. During GO term probability prediction, each feature channel of the labeled features is linearly combined and then passed through a sigmoid function to yield the predicted probability of the protein function.

The final prediction probabilities for GO terms are calculated by averaging the probabilities from Classifier 1 and Classifier 2.

2.2.7 Multi-grained focal loss function

The model aims to predict hundreds or even thousands of GO terms, though each protein generally corresponds to only a small subset of these terms. This leads to a severe imbalance between positive and negative labels during training. When calculating the loss, negative labels often dominate, which affects the model's ability to predict positive labels. Additionally, many GO terms in the training set occur infrequently, leading the model to favor high-frequency terms during training while often overlooking these low-frequency yet more specific GO terms.

Specifically, the focal parameter γ in the traditional focal loss function (Ross and Dollár 2017) is refined into the positive and negative focal parameters $\gamma^{+(i)}$ and $\gamma^{-(i)}$ of each GO_i . Then, parameters $\gamma^{+(i)}$ and $\gamma^{-(i)}$ are further decoupled into the above two granularities (Zhang et al. 2023): the

positive and negative label focus parameter γ_{pn} and the head-tail focus parameter $\gamma_{ht}^{(i)}$. When $\gamma^{+(i)} < \gamma^{-(i)}$, the contribution of negative labels in the loss can be reduced, allowing the model to focus on positive labels. Here, we set $\gamma^{+(i)} = 0$ to maximize the contribution of positive labels in the loss. For $\gamma^{-(i)}$, the sum of γ_{pn}^- and $\gamma_{ht}^{(i)}$ constitutes the focus parameter $\gamma^{-(i)}$ of the negative label GO_i . γ_{pn}^- (Ridnik et al. 2021) is used to adjust the negative focus ratio of all negative labels, while $\gamma_{ht}^{(i)}$ is responsible for adjusting the long-tail focus ratio of the negative label GO_i . $\gamma_{ht}^{(i)}$ adjusts the head-tail label focus using the maximum regularization function (Patro 2015) based on the training set's static distribution of protein functions. As shown in Supplementary Fig. S3, low-frequency tail labels have larger $\gamma_{ht}^{(i)}$ values. A larger $\gamma_{ht}^{(i)}$ value for tail labels enhances the model's sensitivity to these labels during gradient backpropagation, encouraging it to allocate more focus to long-tail labels. We analyze the impact of different $\gamma_{ht}^{(i)}$ values on model performance in the "Sensitivity Analysis" section of the Supplementary material. As shown in Equation 3, the loss function $\mathcal{L}_{mgfl}$ is defined as follows:

$$\mathcal{L}_{mgfl}(p) = \begin{cases} \mathcal{L}_{mgfl}^+ = \sum_{i=1}^C \log(p) \gamma_{ht}^{(i)} & , \text{ if } y = 1 \\ \mathcal{L}_{mgfl}^- = \sum_{i=1}^C p^{1-(i)} \log(1-p) \gamma_{ht}^{(i)} & , \text{ if } y = 0 \end{cases} \quad (3)$$

C is the number of functional categories, i represents the i^{th} protein function, p is the predicted probability, and y is the actual label (0, 1). $\gamma_{ht}^{(i)}$ is defined as follows:

$$\gamma_{ht}^{(i)} = \sqrt{(-\log(go_i/go_{max}) + 1)} \quad (4)$$

go_i is the frequency of occurrence of GO term i , and go_{max} is the maximum frequency of occurrence of all GO terms. α is used to adjust the value of $\gamma_{ht}^{(i)}$.

3 Results and discussion

In order to comprehensively evaluate the performance of GOBoost, we first compared it with other advanced methods on the PDB and AF2 datasets. Then, we showed the specific advantages of GOBoost over the current SOTA HEAL method on the PDB dataset. Finally, we conducted a series of ablation experiments to analyze our designed module's specific improvement effect in depth. We used three evaluation metrics: protein-centric Fmax, Smin, and function-centric area under the precision-recall curve (AUPR), which are commonly used in the CAFA challenge. For the detailed calculation methods of these metrics, please refer to the "Evaluation metrics" section in the Supplementary material.

3.1 Performance on the PDB dataset

As shown in Table 2, GOBoost achieved the best results on all evaluation metrics for the three sub-function ontologies. Specifically, GOBoost achieved scores of 0.765, 0.458, and 0.573 on AUPR, 0.787, 0.659, and 0.745 on Fmax, and 0.292, 0.450, and 0.401 on Smin for MF, BP, and CC functions, respectively. The prediction performance of GOBoost significantly surpassed Blast, FunFams (Das et al. 2015), and DeepGO (Kulmanov et al. 2018). This result shows that relying solely on protein sequence information to predict complex protein functions is not enough. Then, compared with

Table 2. Performance on the PDB dataset.

Method	AUPR (↑)			Fmax (↑)			Smin (↓)		
	MF	BP	CC	MF	BP	CC	MF	BP	CC
GOBoost	0.765	0.458	0.573	0.787	0.659	0.745	0.292	0.450	0.401
GOBoost ^{All}	0.733	0.415	0.546	0.769	0.637	0.739	0.309	0.471	0.408
GOBoost ^{All(PDB)}	0.629	0.311	0.405	0.702	0.574	0.680	0.373	0.517	0.465
Blast	0.136	0.067	0.097	0.038	0.336	0.448	0.632	0.651	0.628
FunFams	0.367	0.260	0.228	0.572	0.500	0.627	0.531	0.579	0.503
DeepGO	0.391	0.182	0.263	0.577	0.493	0.594	0.472	0.577	0.550
DeepFRI	0.495	0.261	0.274	0.625	0.540	0.613	0.437	0.543	0.527
HEAL ^{PDB}	0.571	0.259	0.342	0.691	0.565	0.655	0.401	0.540	0.501
HEAL	0.691	0.337	0.467	0.747	0.595	0.687	0.342	0.509	0.458

Note: Data in bold indicate the best result. GOBoost and GOBoost^{All} are trained on PDB and AF2 training datasets. GOBoost^{All(PDB)} is trained only on the PDB training set. GOBoost^{All} and GOBoost^{All(PDB)} are base models without ensemble strategy.

the latest protein structure-based function prediction method HEAL (Gu *et al.* 2023), GOBoost increased AUPR scores by 10.71%, 35.91%, and 22.71% on the three sub-function ontologies, increased Fmax scores by 5.35%, 10.76%, and 8.44%, and decreased Smin scores by 14.62%, 11.59%, and 12.45%. The BP classification function, which requires the model to predict more GO terms, is more challenging than CC and MF. GOBoost's performance improvement is particularly significant when predicting the BP classification function. Notably, GOBoost has the most significant improvement in AUPR, especially for BP, where it outperformed the HEAL method by 35.91%. It highlights GOBoost's robustness, indicating that it effectively recognizes positive labels across various thresholds, handles imbalanced label data, and is not easily affected by threshold selection.

GOBoost^{All} achieved scores of 0.733, 0.415, and 0.546 on AUPR, 0.769, 0.637, and 0.739 on Fmax, and 0.309, 0.471, and 0.408 on Smin, for the MF, BP, and CC function categories. Compared to the HEAL method, GOBoost^{All} increased AUPR by 6.08%, 23.15%, and 16.92%, respectively, across the three function types. Fmax increased by 2.95%, 7.06%, and 7.57%, and Smin decreased by 9.65%, 7.47%, and 10.92%. Although the performance of GOBoost^{All} was slightly lower than GOBoost, it still outperformed the HEAL method in all evaluation metrics. The experimental data indicate a consistent improvement trend between GOBoost^{All} and GOBoost, with both achieving the best improvement effect on AUPR. Importantly, even without applying the long-tail optimization ensemble strategy proposed in this work, the base model still effectively handles this imbalanced data.

This work further analyzed the performance of our base model GOBoost^{All(PDB)}. GOBoost^{All(PDB)} achieved scores of 0.629, 0.311, and 0.405 on AUPR, 0.702, 0.574, and 0.680 on Fmax, and 0.373, 0.517, and 0.465 on Smin for MF, BP, and CC functions, respectively. GOBoost^{All-PDB} showed superior predictive performance compared to Blast, FunFams, DeepGO, and DeepFRI (Gligorijević *et al.* 2021). Compared to DeepFRI, GOBoost^{All(PDB)} showed an AUPR improvement of 27.07%, 19.16% and 47.81% across MF, BP, and CC, respectively. Fmax increased by 12.32%, 6.30% and 10.93%, and Smin decreased by 14.65%, 4.76%, and 11.76%. Compared with HEAL^{PDB}, GOBoost^{All(PDB)} consistently delivers better results in all evaluation metrics. These results indicate that the GOBoost base model outperforms other methods even when trained on limited amounts of data.

Table 3. Performance on the AF2 dataset.

Method	MF		BP		CC	
	AUPR (↑)	Fmax (↑)	AUPR (↑)	Fmax (↑)	AUPR (↑)	Fmax (↑)
GOBoost	0.582	0.556	0.246	0.497	0.318	0.643
GOBoost ^{All}	0.540	0.526	0.220	0.480	0.313	0.621
GOBoost ^{All(PDB)}	0.366	0.418	0.121	0.397	0.186	0.521
DeepFRI	0.342	0.398	0.114	0.387	0.192	0.536
HEAL ^{PDB}	0.345	0.414	0.104	0.408	0.141	0.513
DeepGOPlus	0.463	0.450	0.203	0.430	0.267	0.567
HEAL	0.502	0.491	0.200	0.475	0.287	0.614

Note: Data in bold indicate the best result. GOBoost and GOBoost^{All} are trained on PDB and AF2 training datasets. GOBoost^{All(PDB)} is trained only on the PDB training set. GOBoost^{All} and GOBoost^{All(PDB)} are base models without ensemble strategy.

3.2 Performance on the AF2 dataset

In the AF2 test set, GOBoost's prediction performance was evaluated under more challenging conditions, where the similarity between the protein sequences in the AF2 test and training sets was less than 25%. To evaluate GOBoost's reliance on sequence similarity, we introduced the DeepGOPlus (Kulmanov and Hoehndorf 2020) method for comparative analysis. DeepGOPlus combines deep learning techniques with transfer annotation based on sequence similarity, which often outperforms traditional methods that rely solely on sequence similarity for annotation transfer.

As shown in Table 3, on the AF2 test set, GOBoost achieved the best results on all evaluation metrics for the three GO categories. GOBoost achieved scores of 0.582, 0.246, and 0.318 on AUPR and 0.556, 0.497, and 0.643 on Fmax for MF, BP, and CC functions. Compared with DeepGOPlus, GOBoost increased AUPR by 25.70%, 21.18%, and 19.10% and Fmax by 23.56%, 15.58%, and 13.40%. In the difficult situation of transferring annotations without high-similarity sequences, GOBoost reduces its dependence on sequence similarity by leveraging structural information and GO terms co-occurrence information. Compared to the HEAL method, GOBoost increased AUPR for three types of functions by 15.64%, 23.00%, and 10.80% and Fmax by 13.24%, 4.63%, and 4.92%. The significant improvement in AUPR further shows GOBoost's ability to recognize positive labels efficiently in difficult situations, making its prediction results robust.

The performance comparison between the GOBoost^{All} and all comparison methods was further analyzed. GOBoost^{All}

achieved 0.540, 0.220, and 0.313 on AUPR and 0.526, 0.480, and 0.313 on Fmax for MF, BP, and CC functions, respectively. GOBoost^{All} consistently outperformed all comparison methods on all evaluation metrics for the three GO categories. This may be because the long-tail strategy we designed prompts the base model to pay more attention to rare functions, thereby improving the overall prediction performance of the model. GOBoost^{All(PDB)} shows that its prediction performance on MF and BP functions surpassed that of DeepFRI, even when DeepFRI was trained with enhanced data. Similar results have also been achieved on CC. GOBoost^{All(PDB)} outperformed HEAL^{PDB} in most evaluation metrics, and only slightly lower than HEAL^{PDB} by 0.011 in Fmax for BP. It could be attributed to the fact that the PDB training set for BP consisted of only 11,295 proteins, limiting the model's ability to learn sufficiently. However, the comparison between GOBoost^{All} and GOBoost^{All(PDB)} revealed that the predictive performance of GOBoost^{All} improved significantly as the amount of trainable protein data increased.

3.3 Performance of GO terms with different IC (information content) values on the PDB test set

We divided the functions into three groups, shallow, normal, and specific, according to their IC values to evaluate the performance of GOBoost in predicting functions of different difficulty levels. The larger the IC value, the higher the specificity value of the function, the lower the frequency of occurrence in the data set, and the more difficult it is to be predicted by the model. The experimental results in Table 4 show that GOBoost and GOBoost^{All} outperformed the HEAL method in all three categories of GO terms. Notably, in the group with the most difficult to predict IC > 10, GOBoost and GOBoost^{All} achieved improvements in AUPR and F1 scores for the three BP, CC, and MF GO categories (52.48%, 16.96%), (31.09%, 12.94%), (13.28%, 9.68%), and (34.71%, 4.09%), (21.76%, 4.85%), (7.56%, 4.22%), respectively. While GOBoost achieved more significant improvement than GOBoost^{All}, likely due to the influence of

ensemble strategy, GOBoost^{All} also achieved considerable improvement in most evaluation groups, especially in the AUPR of IC > 10 group, which increased by 34.71%, 21.76%, and 7.56%. The improvement of GOBoost^{All} in Fmax for the group with IC > 10 is slightly inferior to other metrics, possibly due to the limited adjustment of the long-tail GO term weights, which aimed to optimize the overall performance of the model in this work. The integration of GOBoost^{Tail} effectively addresses this limitation, with GOBoost enhancing the performance over GOBoost^{All} by 0.044, 0.025, and 0.022 respectively.

From the experimental results, it can be found that the prediction performance of GOBoost^{Head} in shallow terms is worse than that of GOBoost^{Tail}, but the prediction performance in specific terms is better than that of GOBoost^{Tail}. GOBoost^{Tail} contains medium-frequency GO terms, GOBoost^{Head} is better than GOBoost^{Tail} on most evaluation indicators of normal terms. The above experimental results show that integrating GOBoost^{Tail} can significantly improve the prediction performance of the model in specific terms, while integrating GOBoost^{Head} can significantly improve the prediction performance of the model in shallow terms. We show and analyze the prediction performance of the model on shallow, normal, and specific terms in the “Example Analysis” section of the [Supplementary material](#).

3.4 P-value and difference analysis of GOBoost and HEAL prediction results on the PDB test set

We performed a P-value analysis to quantify the difference in prediction results between the GOBoost and HEAL methods. We performed the Shapiro–Wilk normality test and found that the data did not conform to a normal distribution. Therefore, we used the Mann–Whitney U test to calculate the P-value. The P-values for MF, BP, and CC categories were of 6.16e-18, 9.70e-28, and 3.01e-08, respectively. There are significant differences in the prediction results of the HEAL method and GOBoost in the three types of functions. In particular, the prediction difference is the largest in the BP function, which is consistent with the previous experimental results. GOBoost has the most significant improvement in the BP function. To further explore the performance improvement of GOBoost over HEAL at different sequence lengths, we divided the proteins into five groups according to their sequence length for analysis. Then, we counted the distribution of the prediction differences between GOBoost and HEAL in each group. It can be seen from [Supplementary Fig. S2](#) that in any interval, the AUPR score difference between GOBoost and HEAL is mostly concentrated in the range of (−0.1, 0.2). However, it is worth noting that the median of these differences always remains at or above 0. Additionally, the interval between the upper quartile and the median is significantly longer than the interval between the lower quartile and the median. Even for those proteins whose prediction results did not improve, the prediction performance of GOBoost was only slightly lower than that of HEAL. In contrast, for those proteins whose prediction results have been improved, the improvement effect of GOBoost is more significant. The average improvement of each group shows that the improvement effect of GOBoost is not affected by the size of the protein graph. GOBoost has the best average improvement effect on BP and CC, even in the largest protein graph group. On MF, GOBoost has an average improvement of 0.074 in the 600–

Table 4. Performance of GO terms with different IC values on the PDB test set.

GO Method		F1 (↑)			AUPR (↑)		
		Shallow	Normal	Specific	Shallow	Normal	Specific
BP	GOBoost:	0.732	0.586	0.400	0.808	0.547	0.369
	GOBoost ^{-Head} :	0.717	0.567	0.399	0.790	0.537	0.369
	GOBoost ^{-Tail} :	0.726	0.565	0.357	0.804	0.512	0.326
	GOBoost ^{-All} :	0.715	0.556	0.356	0.788	0.505	0.326
	HEAL:	0.686	0.525	0.342	0.750	0.428	0.242
CC	GOBoost:	0.409	0.665	0.349	0.833	0.644	0.506
	GOBoost ^{-Head} :	0.409	0.660	0.348	0.815	0.639	0.506
	GOBoost ^{-Tail} :	0.409	0.665	0.324	0.829	0.632	0.470
	GOBoost ^{-All} :	0.409	0.659	0.324	0.812	0.627	0.470
	HEAL:	0.336	0.620	0.309	0.779	0.510	0.386
MF	GOBoost:	0.481	0.761	0.442	0.935	0.815	0.674
	GOBoost ^{-Head} :	0.472	0.745	0.443	0.933	0.809	0.674
	GOBoost ^{-Tail} :	0.479	0.745	0.420	0.934	0.788	0.641
	GOBoost ^{-All} :	0.471	0.738	0.420	0.932	0.783	0.640
	HEAL:	0.464	0.697	0.403	0.917	0.739	0.595

Note: Shallow represents the protein functional group with IC < 5, Normal represents the protein functional group with 5 ≤ IC < 10, and specific represents the protein functional group with IC ≥ 10. Data in bold indicate the best result. GOBoost^{Head} is the model obtained by integrating GOBoost^{Tail} and GOBoost^{All}, without GOBoost^{Head}. GOBoost^{Tail} is the model obtained by integrating GOBoost^{Head} and GOBoost^{All}, without GOBoost^{Tail}.

800 residue range and an average of 0.047 on the largest protein graph group, without too many extreme negative values.

3.5 Ablation study

We performed ablation experiments on different ensemble optimization strategies and various ensemble parts on two datasets. Ablation study results showed that the proposed long-tail optimization integration strategy achieves the best results. For specific ablation study results and analysis, see the [Supplementary material](#) paragraph “Ablation study of long-tail optimization ensemble strategy on the PDB and AF2 DataSet.” We further performed ablation experiments on the “Global-local label graph” submodule and the “Multi-grained focal loss function,” and performed more detailed ablation experiments on their long-tail designs to observe their effects on model performance. Ablation study results showed that these strategies improve the performance of the model overall. For specific ablation study results and analysis, see the [Supplementary material](#) paragraph “Ablation study of long-tail design in the base model on the PDB and AF2 dataset.”

4 Conclusion

In this work, we proposed a new method GOBoost to address the long-tail distribution effect of functional labels. We proposed a long-tail optimization ensemble strategy. Different from previous works that usually integrate deep learning models with non-deep learning models, our work integrates three deep learning models with different training objectives. Furthermore, we designed a global-local label graph module from two perspectives: the global function of the dataset and the local individual function of specific proteins, to [supplement](#) the connection between long-tail labels and other labels. In addition, the $\mathcal{L}_{mgfl}$ loss is designed to appropriately increase the weight of the long-tail GO terms so that the model pays more attention to the long-tail function, thereby improving the overall prediction performance. We evaluated the performance of GOBoost on two datasets and compared it to the SOTA method; it achieved comprehensive improvements on all evaluation metrics for the three GO categories. The best results were achieved in the PDB test set containing more protein samples, indicating that GOBoost is highly robust in predicting the functions of proteins with a variety of different structures. GOBoost achieves the best results on the difficult AF2 test set, showing GOBoost’s capability to generalize and predict challenging proteins.

Designing models from the long-tail perspective has showed great potential in protein function prediction. Looking ahead, we aim to leverage AlphaFold 3 to enhance the accuracy of protein structure predictions for model training. As many protein functions are carried out through the interaction of multiple proteins, incorporating protein interaction network data could provide valuable contextual information, helping infer the biological processes and functions in which these proteins may participate. Additionally, we plan to integrate various non-deep learning methods, leveraging the strengths of each approach to select the most accurate predictions from a range of preliminary results.

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Supplementary data

[Supplementary data](#) are available at *Bioinformatics* online.

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References

- Abdine H, Chatzianastasis M, Bouyioukos C *et al.* Prot2text: Multimodal protein’s function generation with gnn and transformers. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 10757–10765 2024.
- Al-Shahib A, Breitling R, Gilbert D. Feature selection and the class imbalance problem in predicting protein function from sequence. *Appl Bioinformatics* 2005;4:195–203.
- Altschul SF, Gish W, Miller W *et al.* Basic local alignment search tool. *J Mol Biol* 1990;215:403–10.
- Boadu F, Cao H, Cheng J. Combining protein sequences and structures with transformers and equivariant graph neural networks to predict protein function. *Bioinformatics* 2023;39:i318–i325.
- Boadu F, Lee A, Cheng J. Deep learning methods for protein function prediction. *Proteomics* 2025;25:e2300471. page
- Boutet E, Lieberherr D, Tognolli M *et al.* Uniprotkb/swiss-prot: the manually annotated section of the uniprot knowledgebase. In: *Plant Bioinformatics: Methods and Protocols*, Totowa, NJ: Springer, 2007, 89–112.
- Buchfink B, Reuter K, Drost H-G. Sensitive protein alignments at tree-of-life scale using diamond. *Nat Methods* 2021;18:366–8.
- Cao Y, Shen Y. Tale: transformer-based protein function annotation with joint sequence-label embedding. *Bioinformatics* 2021;37:2825–33.
- Dana JM, Gutmanas A, Tyagi N *et al.* Sifts: updated structure integration with function, taxonomy and sequences resource allows 40-fold increase in coverage of structure-based annotations for proteins. *Nucleic Acids Res* 2019;47:D482–9.
- Das S, Lee D, Sillitoe I *et al.* Functional classification of cath superfamilies: a domain-based approach for protein function annotation. *Bioinformatics* 2015;31:3460–7.
- Elnaggar A, Heinzinger M, Dallago C *et al.* Prottrans: toward understanding the language of life through self-supervised learning. *IEEE Trans Pattern Anal Mach Intell* 2021;44:7112–27.

- Giri N, Cheng J. De novo atomic protein structure modeling for cryoem density maps using 3d transformer and hmm. *Nat Commun* 2024; 15:5511.
- Gligorijević V, Renfrew PD, Kosciółek T *et al.* Structure-based protein function prediction using graph convolutional networks. *Nature Communications* 2021;12:3168.
- Gu Z, Luo X, Chen J *et al.* Hierarchical graph transformer with contrastive learning for protein function prediction. *Bioinformatics* 2023; 39:btad410.
- Huntley RP, Sawford T, Mutowo-Meullenet P *et al.* The Goa database: gene ontology annotation updates for 2015. *Nucleic Acids Res* 2015;43:D1057–63.
- Jumper J, Evans R, Pritzel A *et al.* Highly accurate protein structure prediction with alphafold. *Nature* 2021;596:583–9.
- Jung J, Thon MR. Automatic annotation of protein functional class from sparse and imbalanced data sets. In: Data Mining and Bioinformatics: First International Workshop, VDMB 2006, Seoul, Korea, September 11, 2006, Revised Selected Papers 1. Berlin, Heidelberg: Springer, 2006, 65–77.
- Kipf TN, Welling M. Semi-supervised classification with graph convolutional networks. *arXiv Preprint arXiv: 1609.02907* 2016.
- Kulmanov M, Hoehndorf R. Deepgoplus: improved protein function prediction from sequence. *Bioinformatics* 2020;36:422–9.
- Kulmanov M, Khan MA, Hoehndorf R *et al.* Deepgo: predicting protein functions from sequence and interactions using a deep ontology-aware classifier. *Bioinformatics* 2018;34:660–8.
- Lai B, Xu J. Accurate protein function prediction via graph attention networks with predicted structure information. *Brief Bioinform* 2022;23:bbab502.
- Li M, Shi W, Zhang F *et al.* A deep learning framework for predicting protein functions with co-occurrence of go terms. *IEEE/ACM Trans Comput Biol Bioinform* 2023;20:833–42.
- Lin Z, Akin H, Rao R *et al.* Language models of protein sequences at the scale of evolution enable accurate structure prediction. *bioRxiv* 2022.
- Notaro M, Frasca M, Petrini A *et al.* Hemdag: a family of modular and scalable hierarchical ensemble methods to improve gene ontology term prediction. *Bioinformatics* 2021;37:4526–33.
- Reference Genome Group of the Gene Ontology Consortium. The gene ontology's reference genome project: a unified framework for functional annotation across species. *PLoS Computational Biology* 2009;5:e1000431.
- Patro S. Normalization: a preprocessing stage. *arXiv Preprint arXiv: 1503.06462* 2015.
- Ranjan A, Tiwari A, Deepak A. A Sub-sequence based approach to protein function prediction via multi-attention based multi-aspect network. *IEEE/ACM Trans Comput Biol Bioinform* 2023;20:94–105.
- Ridnik T, Ben-Baruch E, Zamir N *et al.* Asymmetric loss for multi-label classification. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 82–91. 2021.
- Rives A, Meier J, Sercu T *et al.* Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proceedings of the National Academy of Sciences* 2021;118: e2016239118.
- Ross T-Y, Dollár G. Focal loss for dense object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 2980–88. 2017.
- Steinegger M, Söding J. Clustering huge protein sequence sets in linear time. *Nat Commun* 2018;9:2542.
- Wang S, Sun S, Li Z *et al.* Accurate de novo prediction of protein contact map by ultra-deep learning model. *PLoS Comput Biol* 2017; 13:e1005324.
- Ye J, He J, Peng X *et al.* Attention-driven dynamic graph convolutional network for multi-label image recognition. In *Computer Vision—ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXI* 16, pp. 649–65. Springer, 2020.
- Yuan Q, Xie J, Xie J *et al.* Fast and accurate protein function prediction from sequence through pretrained language model and homology-based label diffusion. *Brief Bioinform* 2023;24:bbad117.
- Zhang W, Liu C, Zeng L *et al.* Learning in imperfect environment: Multi-label classification with long-tailed distribution and partial labels. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 1423–32. 2023.
- Zhao C, Liu T, Wang Z. Panda2: protein function prediction using graph neural networks. *NAR Genom Bioinform* 2022; 4:lqac004.
- Zhao C, Liu T, Wang Z. Panda-3d: protein function prediction based on alphafold models. *NAR Genom Bioinform* 2024;6:lqae094.
- Zhou B, Khosla A, Lapedriza A *et al.* Learning deep features for discriminative localization. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 2921–9. 2016.